Difference of Convex Functions in Robust MPC

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Interplay Between Machine Learning and Set-Based Identification & Control



Workshop at the American Control Conference 2025
Monday, July 7th, 2025. Denver, CO.

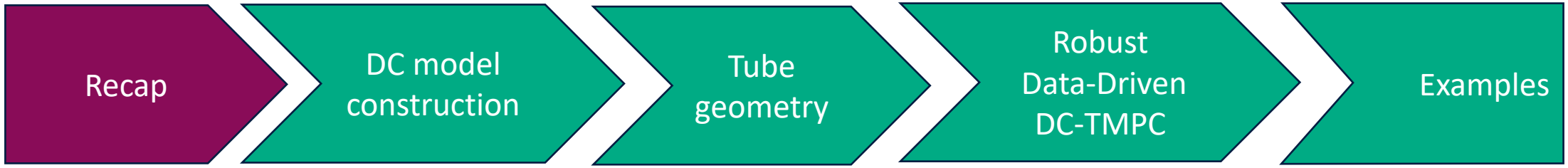
7:50 am – 8:00 am	Opening Remarks
8:00 am – 8:45 am	*Difference of Convex Functions in Robust Data-Driven MPC: Part 1* **Yana Lishkova**, University of Oxford
8:45 am – 9:35 am	*Difference of Convex Functions in Robust Data-Driven MPC: Part 2* **Mark Cannon**, University of Oxford
9:35 am – 10:00 am	Coffee Break 1
10:00 am – 10:50 am	*Data-driven Computation of Robust Control Invariant Sets with Concurrent Model Selection* **Necmiye Ozay**, University of Michigan
10:50 am – 11:40 am	*Safety-Critical Control through Massive Parallel Simulation* **Aaron D. Ames**, California Institute of Technology
11:40 am – 12:30 pm	*Why Do We Need "Control" in Control-Oriented Learning?* **Mario Sznaier**, Northeastern University
12:30 pm – 1:45 pm	Lunch Break
1:45 pm – 2:45 pm	*Online Uncertainty Set Prediction and Set-Membership Estimation* **Heng Yang**, Harvard University
2:45 pm – 3:30 pm	*Recent Advances on Set Membership Estimation Through the Lens of Statistical Learning* **Yingying Li**, University of Illinois Urbana-Champaign
3:30 pm – 3:50 pm	Coffee Break 2
3:50 pm – 4:40 pm	*Set-based and Machine Learning Perspectives on Control-Affine Dynamics* **Sarah Dean**, Cornell University
4:40 pm – 5:30 pm	*Adversarial Learning-based Online Control for Time-varying and Large-scale Systems* **Jing Yu**, University of Washington

Talk overview

Talk Part 1:



Talk Part 2:



Motivation

- Optimal control:
  - Obtain a feasible control sequence that optimizes a given performance index, subject to constraints.

- Standard problem:

$$\min_{\mathbf{u},\,\mathbf{x}} \sum_{i=0}^{N} \left(\|x_i - x_k^r\|_Q^2 + \|u_i - u_i^r\|_R^2 \right)$$

$$\text{subject to, for } i = 1, \ldots, N-1$$

$$x_0 = x_{\text{init}}$$

$$x_{i+1} = f(x_i, u_i)$$

$$x_i \in \mathcal{X}, \quad u_i \in \mathcal{U}$$

where $\mathbf{u} = \{u_0, \ldots, u_{N-1}\}$, $\mathbf{x} = \{x_0, \ldots, x_N\}$.

Problem 1 - no feedback

$$\min_{\mathbf{u},\,\mathbf{x}} \sum_{i=0}^{N} \left(\|x_i - x_k^r\|_Q^2 + \|u_i - u_i^r\|_R^2 \right)$$

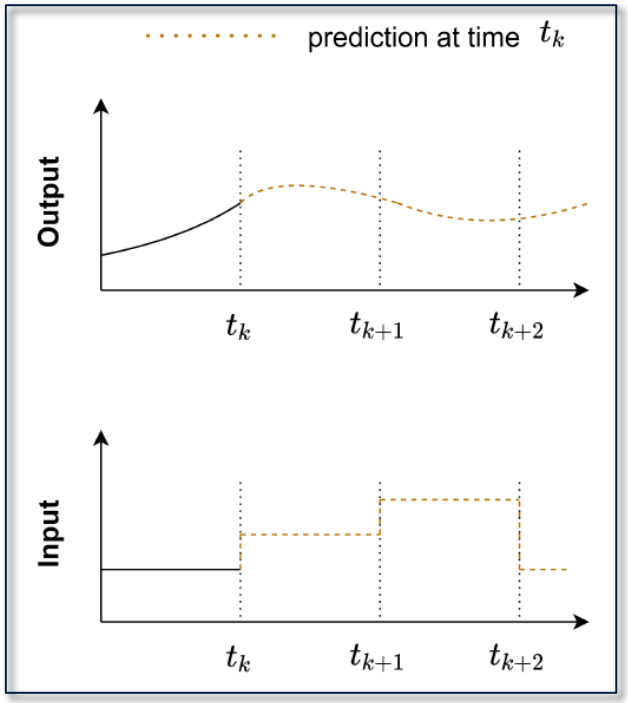
subject to, for $i = 1, \ldots, N-1$

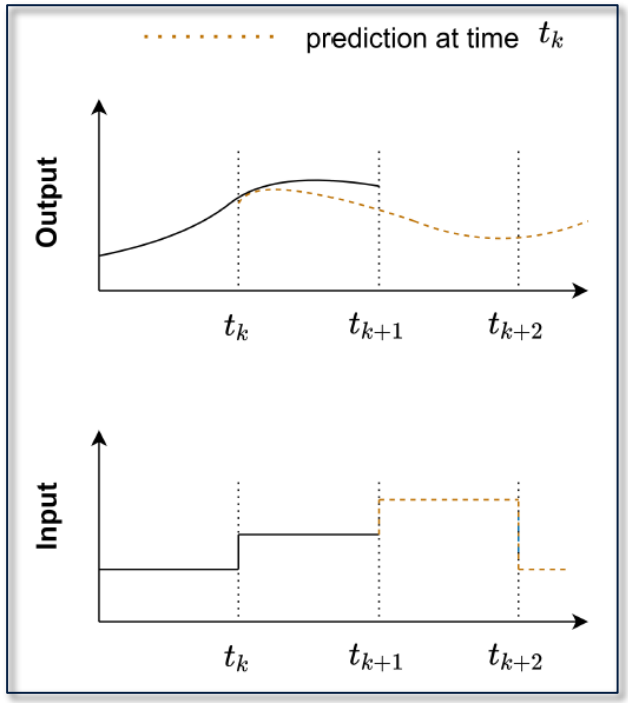
$$x_0 = x_{\text{init}}$$

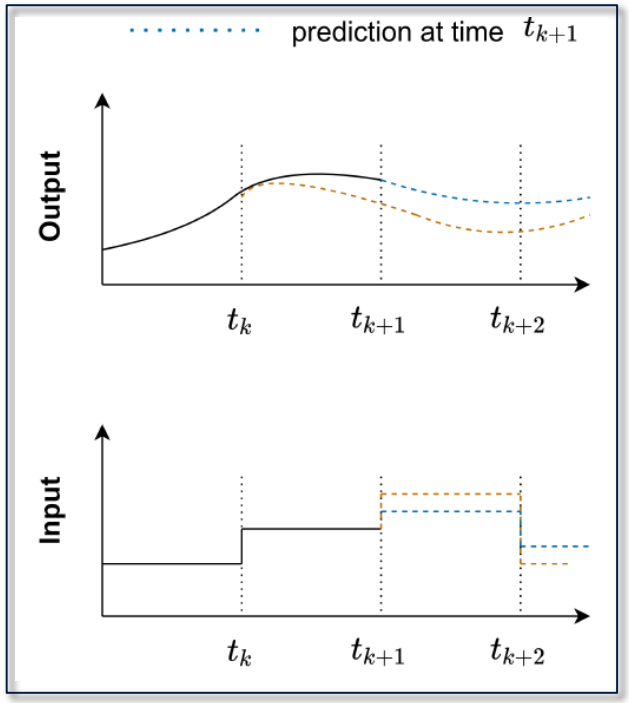
$$x_{i+1} = f(x_i, u_i)$$

$$x_i \in \mathcal{X}, \quad u_i \in \mathcal{U}$$

- What happens when we have:
  - Disturbances
  - Model inaccuracies
  - Changing environment
- **Model Predictive control**







Problem 2 – no time to waste

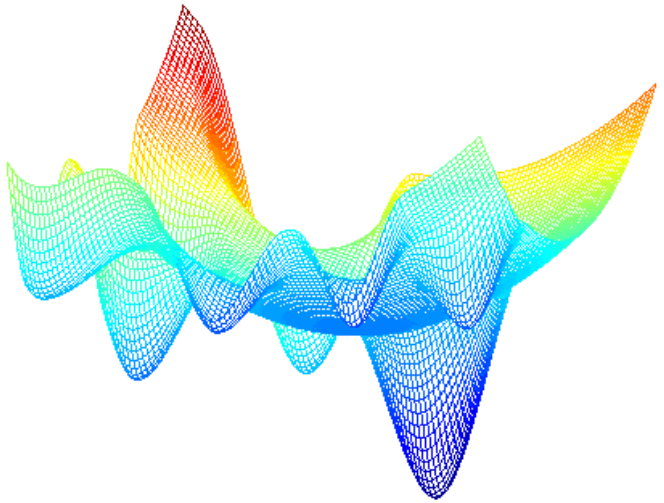
$$\min_{\mathbf{u},\,\mathbf{x}} \sum_{i=0}^{N} \left(\|x_i - x_k^r\|_Q^2 + \|u_i - u_i^r\|_R^2 \right)$$

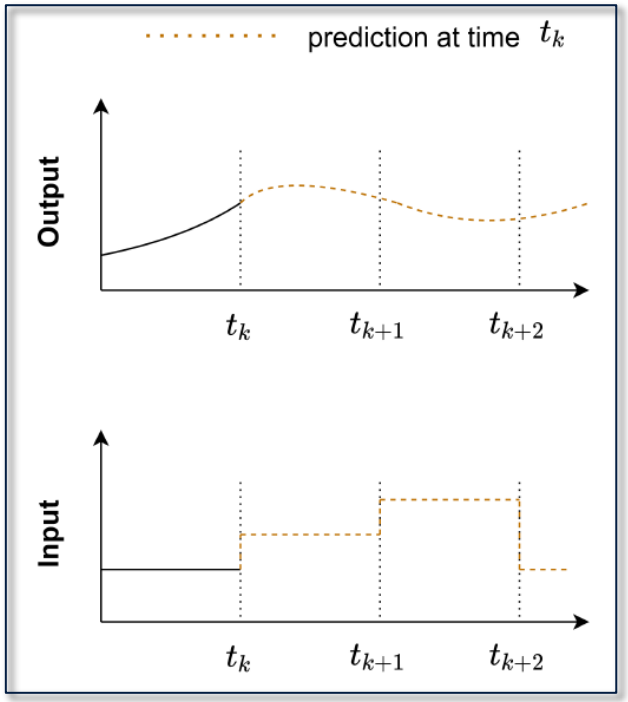
subject to, for $i = 1, \ldots, N-1$

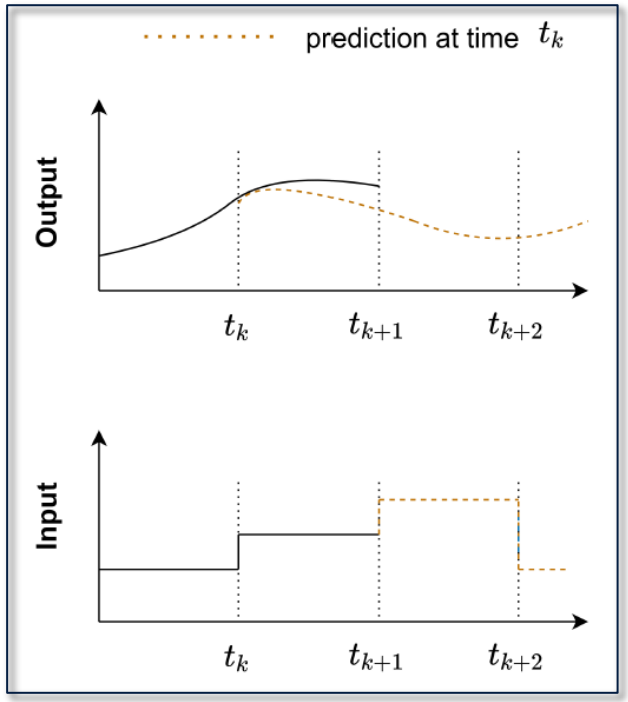
$x_0 = x_{\text{init}}$

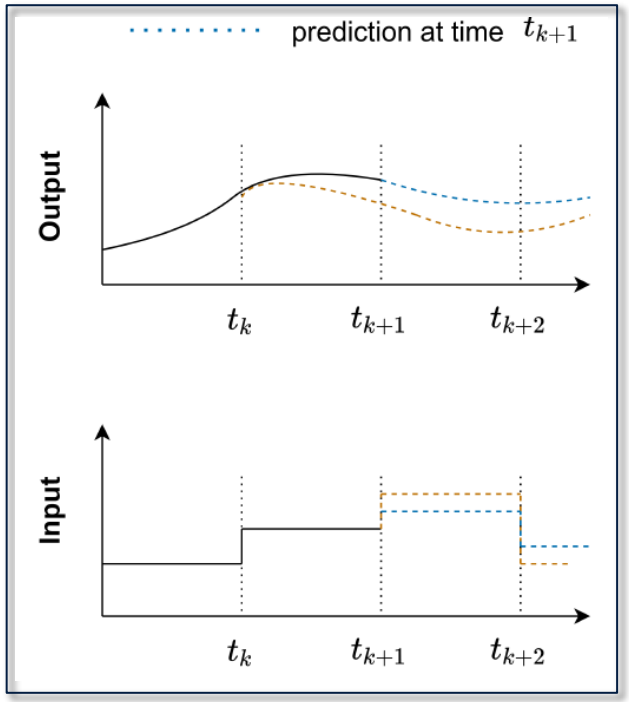
$x_{i+1} = f(x_i, u_i),$

$x_i \in \mathcal{X}, \quad u_i \in \mathcal{U}$

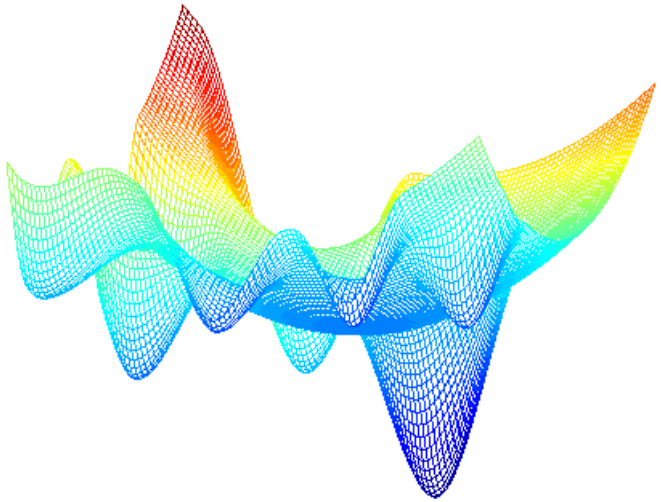








Iterative approximation techniques



- Rely on sequential problem approximations
- Iterative convergence to or close to the true (local) optimum
- Possibility of early termination

Typical approach: **Sequential convex programming**

applied to a nonconvex problem, e.g.

$$\begin{aligned} \text{minimize} \quad & f_0(x) \\ \text{s.t.} \quad & f_i(x) \leq 0, \quad i = 1, \ldots, m \\ & r_i(x) = 0 \quad i = 1, \ldots, p \end{aligned}$$

f_0, f_i nonconvex

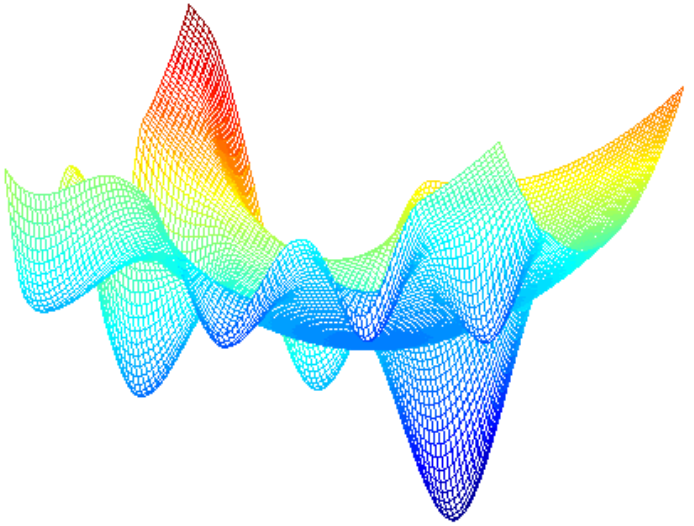
r_i non-affine

Sequential convex programming

- At iter ***k*** , solution estimate $x^{(k+1)}$ obtained using a trust region $\mathcal{T}^{(k)}$ (typically a box)

$$\mathcal{T}^{(k)} = \left\{x : \left|[x]_i - [x^{(k)}]_i\right| \leq \rho_i \quad i = 1, \ldots, n\right\}$$

- $\hat{f}_i(x)$ convex approximation of $f_i(x)$ on $\mathcal{T}^{(k)}$
- $\hat{r}_i(x)$ linear approximation of $r_i(x)$ on $\mathcal{T}^{(k)}$



- $x^{(k+1)}$ is the solution of

$$\begin{aligned} \text{minimize} \quad & \hat{f}_0(x) \\ \text{s.t.} \quad & \hat{f}_i(x) \leq 0, \qquad i = 1, \ldots, m \\ & \hat{h}_i(x) = 0, \qquad i = 1, \ldots, p \\ & x \in \mathcal{T}^{(k)} \end{aligned}$$

- Then update the trust region using heuristics, for example:

$$\rho_k = \frac{f_0(x^k) - f_0(x^{k+1})}{\hat{f}_0^k(x^k) - \hat{f}_0^k(x^{k+1})}$$

Sequential convex programming (SCP):

- A sequence of convex approximations to the original problem
- Relies on ***local model approximations*** and ***trust regions*** for validity
  - Often heuristic and overly conservative
  - Can lead to infeasibility or slow convergence
  - Limited progress per iteration

DC programming

- Any $f \in \mathcal{C}^2$ can be expressed as $f = g - h$ for convex g, h

$$\begin{aligned} \text{minimize} \quad & f_0(x) \\ \text{s.t.} \quad & f_i(x) \leq 0, \quad i = 1, \ldots, m \end{aligned}$$

- Reformulate the problem using differences of convex (DC) functions

$$\begin{aligned} \text{minimize} \quad & g_0(x) - h_0(x) \\ \text{s.t.} \quad & g_i(x) - h_i(x) \leq 0, \quad i = 1, \ldots, m \end{aligned}$$

$f_i = g_i - h_i, \quad g_i, h_i \quad \text{convex } \forall i$

This is a **DC**-program

DC programming

$$\begin{aligned} \text{minimize} \quad & g_0(x) - h_0(x) \\ \text{s.t.} \quad & g_i(x) - h_i(x) \leq 0, \quad i = 1, \ldots, m \end{aligned}$$

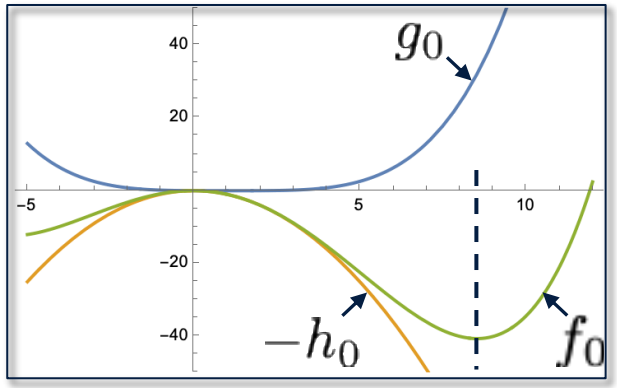
- Next apply **the convex-concave procedure**

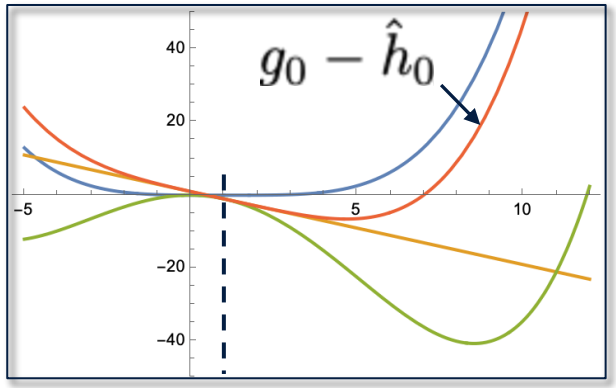
At iteration $k+1$ approximate the concave part based on solution x^k: $\hat{h}_i(x) = h_i(x^k) + \dfrac{\partial h_i}{\partial x}(x^k)(x - x^k)$

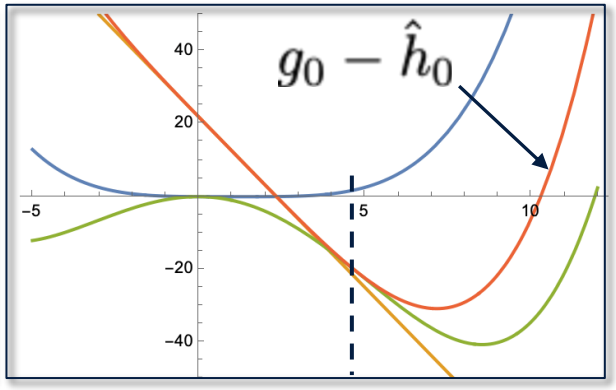
$$\begin{aligned} x^{k+1} = \arg\min \quad & g_0(x) - \hat{h}_0(x) \\ \text{s.t.} \quad & g_i(x) - \hat{h}_i(x) \leq 0, \quad i = 1, \ldots, m \end{aligned}$$

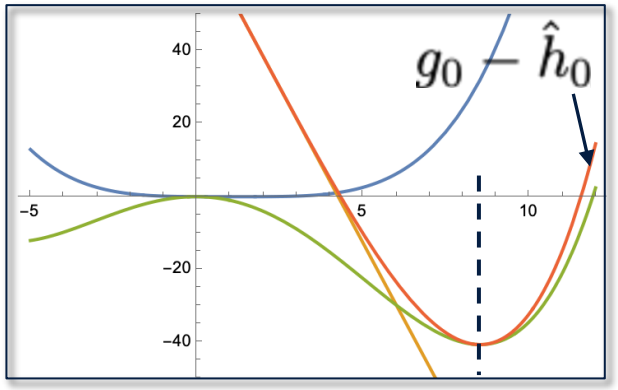
$$\begin{aligned} &\text{as } g_i(x^k) - \hat{h}(x^k) \leq 0 && \Rightarrow g_i(x^k) - h_i(x^k) = f_i(x^k) \leq 0 \ \forall k \text{ and } x^k \to x^*_{\text{NLP}} \\ &\text{and } \hat{h}(x^k) \leq h(x^k) && \Rightarrow \text{Trust region is not needed} \end{aligned}$$

DC programming









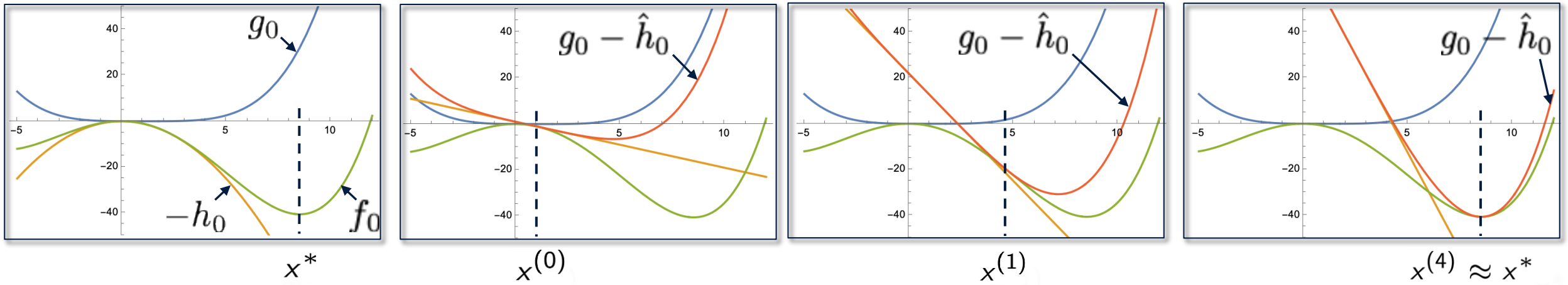
$$x^{k+1} = \arg\min \; g_0(x) - \hat{h}_0(x)$$
$$\text{s.t.} \quad g_i(x) - \hat{h}_i(x) \leq 0, \quad i = 1, \ldots, m$$

as $g_i(x^k) - \hat{h}(x^k) \leq 0$ and $\hat{h}(x^k) \leq h(x^k)$

$\Rightarrow g_i(x^k) - h_i(x^k) = f_i(x^k) \leq 0 \; \forall k$ and $x^k \to x^*_{\text{NLP}}$

$\Rightarrow$ Trust region is not needed

DC programming



- Trust region is not needed because $\hat{f}(x) \geq f(x)\ \forall x$
- Therefore
  - ★ $\hat{f}_0(x^{(k+1)})$ is an upper bound on the true objective $f_0(x^{(k+1)})$
  - ★ the true feasible set contains the feasible set of the approximation

Convex-concave vs SCP

Sequential convex programming (SCP):

- A sequence of convex approximations to the original problem
- Relies on ***local model approximations*** and ***trust regions*** for validity
  - Often heuristic and overly conservative
  - Can lead to infeasibility or slow convergence
  - Limited progress per iteration

Convex-concave procedure (and DC programming)

- Retains full information on the convex part
- Uses a ***global*** overestimator $g_i(x) - \hat{h}_i(x)$ → no progress limit
- Applicable for ***all continuous models*** -> *g, h* obtained using ***inference***
- Proven success in MPC with optimality and stability guarantees

Talk overview



How can it be used in MPC

$$\min_{\boldsymbol{u}_i,\, \boldsymbol{x}_i} \|x_{N|i} - x^r_{N+i}\|^2_{Q_N}$$
$$+ \sum_{k=0}^{N-1} \left(\|x_{k|i} - x^r_{k+i}|^2_Q + \|u_{k|i} - u^r_{k+i}\|^2_R \right)$$

subject to, for $k = 0, \ldots, N-1$:

$$x_{k|i} \in \mathcal{X}$$

$$u_{k|i} \in \mathcal{U}$$

$$x_{k+1|i} = f(x_{k|i}, u_{k|i})$$

$$x_{0|i} = x_i$$

We assume:

- $f : (\mathcal{X}, \mathcal{U}) \to \mathbb{R}^{n_x}$ is differentiable and componentwise convex
- the provision of an initial seed trajectory $(\mathbf{x}^0, \mathbf{u}^0)$ which is feasible

$\boldsymbol{u}_i := \{u_{0|i}, \ldots, u_{N-1|i}\}$, $\boldsymbol{x}_i := \{x_{0|i}, \ldots, x_{N|i}\}$
$x_{k|i}$, $u_{k|i}$- the state and control input at time $k + i$ predicted at time i.

Convex formulation

$$\min_{u_i,\, x_i} \Bigg(\max_{x_{N|i} \in X_{N|i}} \|x_{N|i} - x^r_{N+i}\|^2_{Q_N} + \sum_{k=0}^{N-1} \max_{x_{k|i} \in X_{k|i}} \left(\|x_{k|i} - x^r_{k+i}\|^2_Q + \|u_{k|i} - u^r_{k+i}\|^2_R \right) \Bigg)$$

subject to, for $k = 0, \ldots, N-1$:

$X_{k|i} \subseteq \mathcal{X}$

$u_{k|i} \in \mathcal{U}$

$X_{k+1|i} = A_{k|i} X_{k|i} \oplus \{B_{k|i} u_{k|i}\} \oplus W_{k|i}$

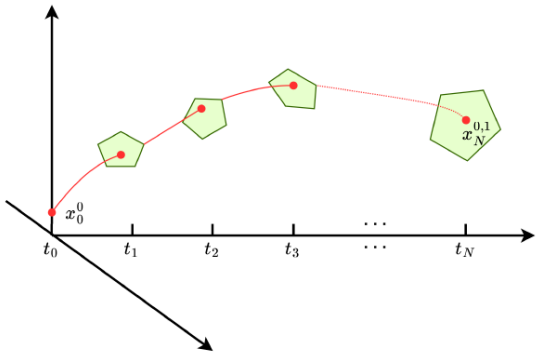
$x_{0|i} = x_i$

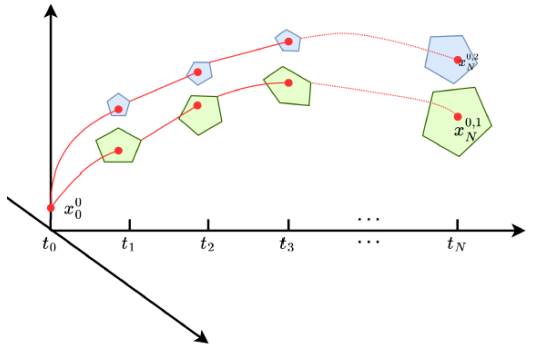
$X_{N|i} \subseteq \mathcal{X}_N \subseteq \mathcal{X}$

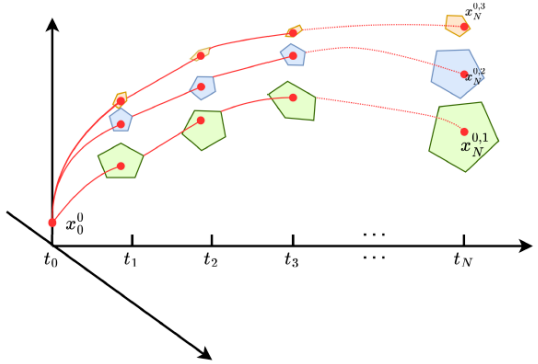
(x^0_i, u^0_i) - seed trajectory

$X_{k|i}$ - tube cross section

$A_{k|i} = \frac{\partial f}{\partial x_{k|i}}|_{(x^0_{k|i}, u^0_{k|i})}, B_{k|i} = \frac{\partial f}{\partial u_{k|i}}|_{(x^0_{k|i}, u^0_{k|i})}$







Convex formulation

$$\min_{u_i,\, x_i} \Bigg(\max_{x_{N|i} \in X_{N|i}} \|x_{N|i} - x^r_{N+i}\|^2_{Q_N} + \sum_{k=0}^{N-1} \max_{x_{k|i} \in X_{k|i}} \left(\|x_{k|i} - x^r_{k+i}\|^2_Q + \|u_{k|i} - u^r_{k+i}\|^2_R \right) \Bigg)$$

subject to, for $k = 0, \ldots, N-1$:

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$$u_{k|i} \in \mathcal{U}$$

$$X_{k+1|i} = A_{k|i} X_{k|i} \oplus \{B_{k|i} u_{k|i}\} \oplus W_{k|i}$$

$$x_{0|i} = x_i$$

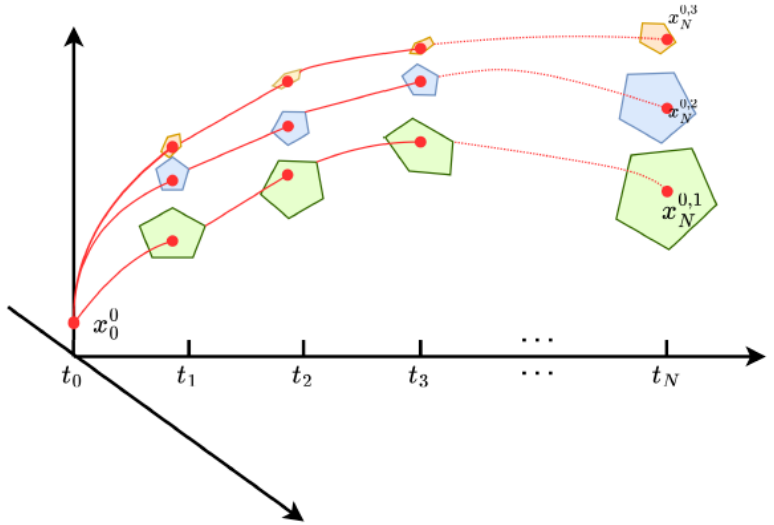
$$X_{N|i} \subseteq \mathcal{X}_N \subseteq \mathcal{X}$$

(x^0_i, u^0_i) - seed trajectory

$X_{k|i}$ - tube cross section

$A_{k|i} = \frac{\partial f}{\partial x_{k|i}}|_{(x^0_{k|i}, u^0_{k|i})}, B_{k|i} = \frac{\partial f}{\partial u_{k|i}}|_{(x^0_{k|i}, u^0_{k|i})}$

- requires heuristic to define the linearization error and shrink the tubes
- conservative- no guarantee of convergence to a local optimal of the original NLP



TMPC

$$\min_{\boldsymbol{u}_i,\, \boldsymbol{x}_i} \Bigg(\max_{x_{N|i} \in X_{N|i}} \|x_{N|i} - x^r_{N+i}\|^2_{Q_N} + \sum_{k=0}^{N-1} \max_{x_{k|i} \in X_{k|i}} \left(\|x_{k|i} - x^r_{k+i}\|^2_Q + \|u_{k|i} - u^r_{k+i}\|^2_R \right) \Bigg)$$

subject to, for $k = 0, \ldots, N-1$:

$X_{k|i} \subseteq \mathcal{X}$

$KX_{k|i} \oplus \{c_{k|i} + c^0_{k|i}\} \subseteq \mathcal{U}$

$X_{k+1|i} = A_{k|i} X_{k|i} \oplus \{B_{k|i} u_{k|i}\} \oplus W_{k|i}$

$x_{0|i} = x_i$

$X_{N|i} \subseteq \mathcal{X}_N \subseteq \mathcal{X}$

$(\boldsymbol{x}^0_i, \boldsymbol{u}^0_i)$ - seed trajectory

$X_{k|i}$ - tube cross section

$A_{k|i} = \frac{\partial f}{\partial x_{k|i}}|_{(x^0_{k|i}, u^0_{k|i})}, B_{k|i} = \frac{\partial f}{\partial u_{k|i}}|_{(x^0_{k|i}, u^0_{k|i})}$

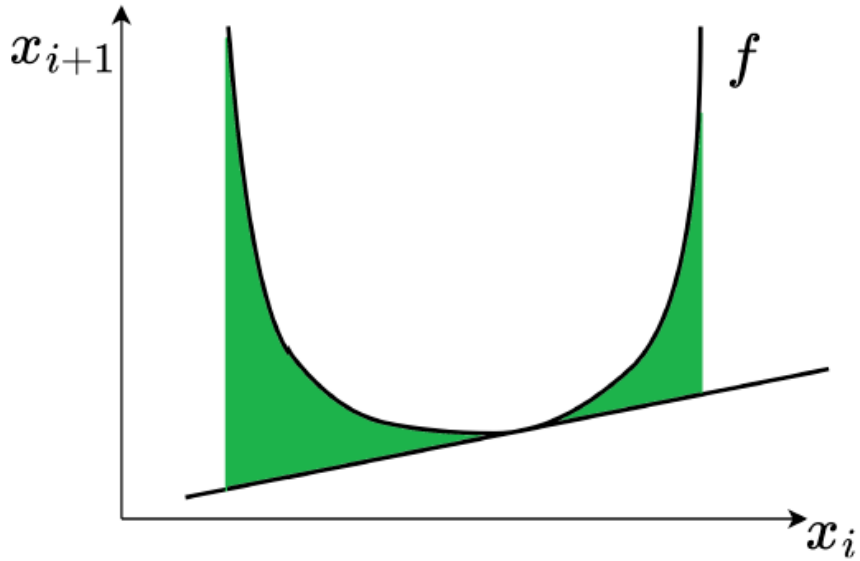
$$[\bar{x}_{k+1|i}]_j \geq [f(x, Kx + c_{k|i} + c^0_{k|i})]_j$$

$$[\underline{x}_{k+1|i}]_j \leq [f(x^0_{k|i,j}, u^0_{k|i,j})]_j + [A_{k|i} + B_{k|i}K]_j (x - x^0_{k|i,j}) + [B_{k|i}]_j c_{k|i}$$

for all $x \in X_{k|i},\ j = 1, \ldots, n_x$

$X_{k|i+1} = \{x : \underline{x}_{k|i+1} \leq x \leq \bar{x}_{k|i+1}\}$

$\boldsymbol{X}_i := \{X_{0|i}, \ldots, X_{N-1|i}\},$



TMPC

A typical problem takes the form:

$$\min_{\boldsymbol{u}_i,\, \boldsymbol{x}_i} \left(\max_{x_{N|i} \in X_{N|i}} \|x_{N|i} - x^r_{N+i}\|^2_{Q_N} + \sum_{k=0}^{N-1} \max_{x_{k|i} \in X_{k|i}} \left(\|x_{k|i} - x^r_{k+i}\|^2_Q + \|u_{k|i} - u^r_{k+i}\|^2_R \right) \right)$$

subject to, for $k = 0, \ldots, N-1$:

$$X_{k|i} \subseteq \mathcal{X}$$

$$KX_{k|i} \oplus \{c_{k|i} + c^0_{k|i}\} \subseteq \mathcal{U}$$

$$X_{k+1|i} = A_{k|i} X_{k|i} \oplus \{B_{k|i} u_{k|i}\} \oplus W_{k|i}$$

$$x_{0|i} = x_i$$

$$X_{N|i} \subseteq \mathcal{X}_N \subseteq \mathcal{X}$$

$(\boldsymbol{x}^0_i, \boldsymbol{u}^0_i)$ - seed trajectory

$X_{k|i}$ - tube cross section

$A_{k|i} = \frac{\partial f}{\partial x_{k|i}}\big|_{(x^0_{k|i}, u^0_{k|i})}, B_{k|i} = \frac{\partial f}{\partial u_{k|i}}\big|_{(x^0_{k|i}, u^0_{k|i})}$

$$[\bar{x}_{k+1|i}]_j \geq [f(x, Kx + c_{k|i} + c^0_{k|i})]_j$$

$$[\underline{x}_{k+1|i}]_j \leq [f(x^0_{k|i,j}, u^0_{k|i,j})]_j + [A_{k|i} + B_{k|i}K]_j (x - x^0_{k|i,j}) + [B_{k|i}]_j c_{k|i}$$

for all $x \in X_{k|i},\ j = 1, \ldots, n_x$

- **At each *k*:**

  While *iter i < maxiter* or $\|\mathbf{c}^*_i\| \geq tol$ **:**

  1. convexify the problem
  2. solve the problem
  3. update the seed x0, c0

- **At each *k*:**
  - Problem is feasible at each iteration $i = 0, 1 \ldots$
  - $J^{(i)}_t \to J^*_{\mathrm{NLP}}$ as $i \to \infty$
  - $\mathbf{X}^{(i)} \to \mathbf{x}^*_{\mathrm{NLP}}$ as $i \to \infty$

What happens is when the assumption is no longer valid

$$\min_{\boldsymbol{u}_i,\, \boldsymbol{x}_i} \|x_{N|i} - x^r_{N+i}\|^2_{Q_N}$$
$$+ \sum_{k=0}^{N-1} \left(\|x_{k|i} - x^r_{k+i}|^2_Q + \|u_{k|i} - u^r_{k+i}\|^2_R \right)$$

subject to, for $k = 0, \ldots, N-1$:

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$$x_{k+1|i} = f(x_{k|i}, u_{k|i})$$

$$x_{0|i} = x_i$$

We assume:

- $f(x,u) = g(x,u) - h(x,u)$, where $g, h : \mathcal{X} \times \mathcal{U} \to \mathbb{R}^{n_x}$ are differentiable an componentwise convex on the convex sets $\mathcal{X}, \mathcal{U}$
- the provision of an initial seed trajectory $(\mathbf{x}^0, \mathbf{u}^0)$ which is feasible problem (but potentially suboptimal)

$\boldsymbol{u}_i := \{u_{0|i}, \ldots, u_{N-1|i}\}$, $\boldsymbol{x}_i := \{x_{0|i}, \ldots, x_{N|i}\}$
$x_{k|i}$, $u_{k|i}$- the state and control input at time $k+i$ predicted at time i.

DC-TMPC convex program

$$\min_{\boldsymbol{u}_i, \boldsymbol{X}_i} \left(\max_{x_{N|i}} \|x_{N|i} - x^r_{N+i}\|^2_{Q_N} + \right.$$

$$\left. \sum_{k=0}^{N-1} \max_{x_{k|i}} \left[\|x_{k|i} - x^r_{k+i}\|^2_Q + \|u_{k|i} - u^r_{k+i}\|^2_R \right] \right)$$

subject to, for $k = 0, \ldots, N-1$:

$$X_{k|i} \in \mathcal{X}$$

$$KX_{k|i} \oplus \{c_{k|i} + c^0_{k|i}\} \subseteq \mathcal{U},$$

$$[\bar{x}_{k+1|i}]_j \geq [f(x, Kx + c_{k|i} + c^0_{k|i})]_j$$

$$[\underline{x}_{k+1|i}]_j \leq [f(x^0_{k|i,j}, u^0_{k|i,j})]_j +$$

$$[A_{k|i} + B_{k|i}K]_j (x - x^0_{k|i,j}) + [B_{k|i}]_j c_{k|i}$$

for all $x \in X_{k|i}$, $j = 1, \ldots, n_x$

$$X_{0|i} = \{x_i\}$$

$$X_{N|i} \in \mathcal{X}_N \subseteq \mathcal{X}$$

$$[\bar{x}_{k+1|i}]_j \geq [g(x, Kx + c_{k|i} + c^0_{k|i})]_j$$
$$- [h(x^{0,h}_{k|i,j}, u^{0,h}_{k|i,j})]_j - [A^h_{k|i} + B^h_{k|i}K]_j (x - x^{0,h}_{k|i,j}) - [B^h_{k|i}]_j c_{k|i}$$

$$[\underline{x}_{k+1|i}]_j \leq [g(x^{0,g}_{k|i,j}, u^{0,g}_{k|i,j})]_j + [A^g_{k|i} + B^g_{k|i}K]_j (x - x^{0,g}_{k|i,j}) + [B^g_{k|i}]_j c_{k|i}$$
$$- [h(x, Kx + c_{k|i} + c^0_{k|i})]_j$$

$$f(x, u) = g(x, u) - h(x, u)$$

DC-TMPC convex program

$$\min_{\boldsymbol{u}_i, \boldsymbol{X}_i} \left(\max_{x_{N|i}} \|x_{N|i} - x^r_{N+i}\|^2_{Q_N} + \right.$$

$$\left. \sum_{k=0}^{N-1} \max_{x_{k|i}} \left[\|x_{k|i} - x^r_{k+i}\|^2_Q + \|u_{k|i} - u^r_{k+i}\|^2_R \right] \right)$$

subject to, for $k = 0, \ldots, N-1$:

$$X_{k|i} \in \mathcal{X}$$

$$KX_{k|i} \oplus \{c_{k|i} + c^0_{k|i}\} \subseteq \mathcal{U},$$

$$[\bar{x}_{k+1|i}]_j \geq [f(x, Kx + c_{k|i} + c^0_{k|i})]_j$$

$$[\underline{x}_{k+1|i}]_j \leq [f(x^{0}_{k|i,j}, u^{0}_{k|i,j})]_j +$$

$$[A_{k|i} + B_{k|i}K]_j (x - x^{0}_{k|i,j}) + [B_{k|i}]_j c_{k|i}$$

for all $x \in X_{k|i}$, $j = 1, \ldots, n_x$

$$X_{0|i} = \{x_i\}$$

$$X_{N|i} \in \mathcal{X}_N \subseteq \mathcal{X}$$

$$[\bar{x}_{k+1|i}]_j \geq [g(x, Kx + c_{k|i} + c^0_{k|i})]_j$$
$$- [h(x^{0,h}_{k|i,j}, u^{0,h}_{k|i,j})]_j - [A^h_{k|i} + B^h_{k|i}K]_j (x - x^{0,h}_{k|i,j}) - [B^h_{k|i}]_j c_{k|i}$$

$$[\underline{x}_{k+1|i}]_j \leq [g(x^{0,g}_{k|i,j}, u^{0,g}_{k|i,j})]_j + [A^g_{k|i} + B^g_{k|i}K]_j (x - x^{0,g}_{k|i,j}) + [B^g_{k|i}]_j c_{k|i}$$
$$- [h(x, Kx + c_{k|i} + c^0_{k|i})]_j \quad \|\mathbf{c}^*_i\| \geq tol$$

- **At each *k*:**

  While *iter i < maxiter* or $\|\mathbf{c}^*_i\| \geq tol$ **:**

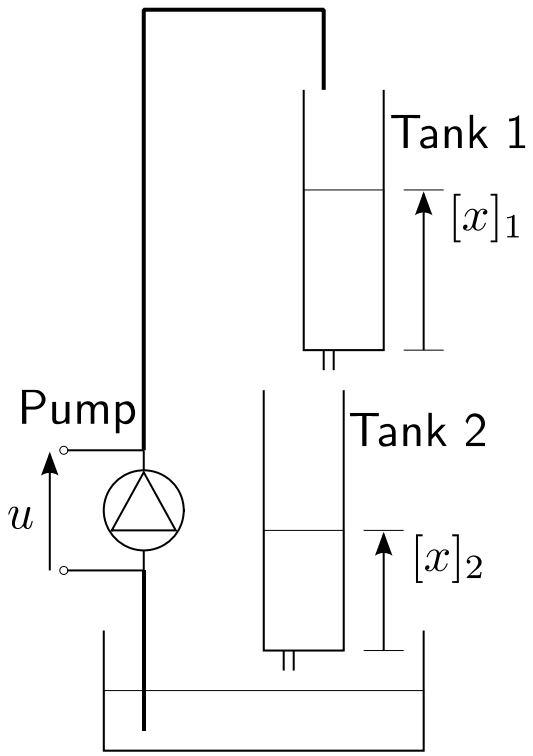
  1. convexify the problem
  2. solve the problem
  3. update the seed x0, c0

- **At each *k*:**
  - Problem is feasible at each iteration $i = 0, 1 \ldots$
  - $J^{(i)}_t \to J^*_{\mathrm{NLP}}$ as $i \to \infty$
  - $\mathbf{X}^{(i)} \to \mathbf{x}^*_{\mathrm{NLP}}$ as $i \to \infty$

Talk overview



Example 1: level control in coupled tanks



- Coupled tanks discrete time model with sampling interval Δ:

$$[x_{t+1}]_1 = [x_t]_1 - \Delta \frac{A_1}{A} \sqrt{2g[x_t]_1} + \Delta \frac{k_p}{A} u_t$$

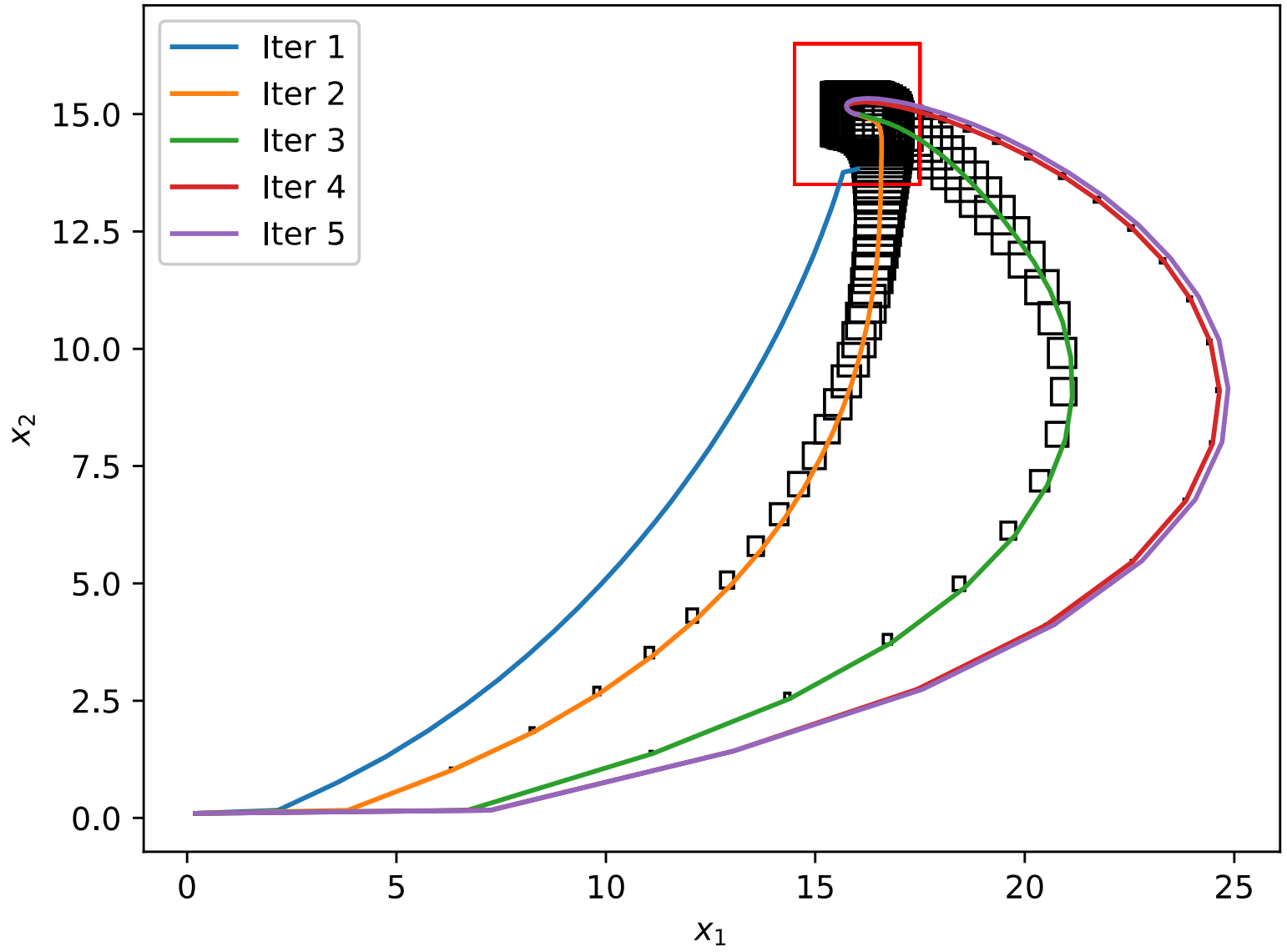
$$[x_{t+1}]_2 = [x_t]_2 - \Delta \frac{A_2}{A} \sqrt{2g[x_t]_2} - \Delta \frac{A_1}{A} \sqrt{2g[x_t]_1}$$

where $[x]_1$, $[x]_2$ are the depths of fluid in tanks 1 and 2, and A, A_1, A_2, g, k_p are parameters

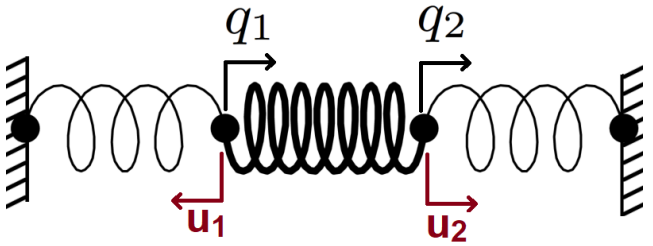
- The model is in the form of a difference of convex functions

$$f(x,u) = \underbrace{\begin{bmatrix} [x]_1 - \Delta \frac{A_1}{A} \sqrt{2g[x]_1} + \Delta \frac{k_p}{A} u \\ [x]_2 - \Delta \frac{A_2}{A} \sqrt{2g[x]_2} \end{bmatrix}}_{=g(x,u)} - \underbrace{\begin{bmatrix} 0 \\ -\Delta \frac{A_1}{A} \sqrt{2g[x]_1} \end{bmatrix}}_{=h(x,u)}$$

Example 1: sequential convex program convergence



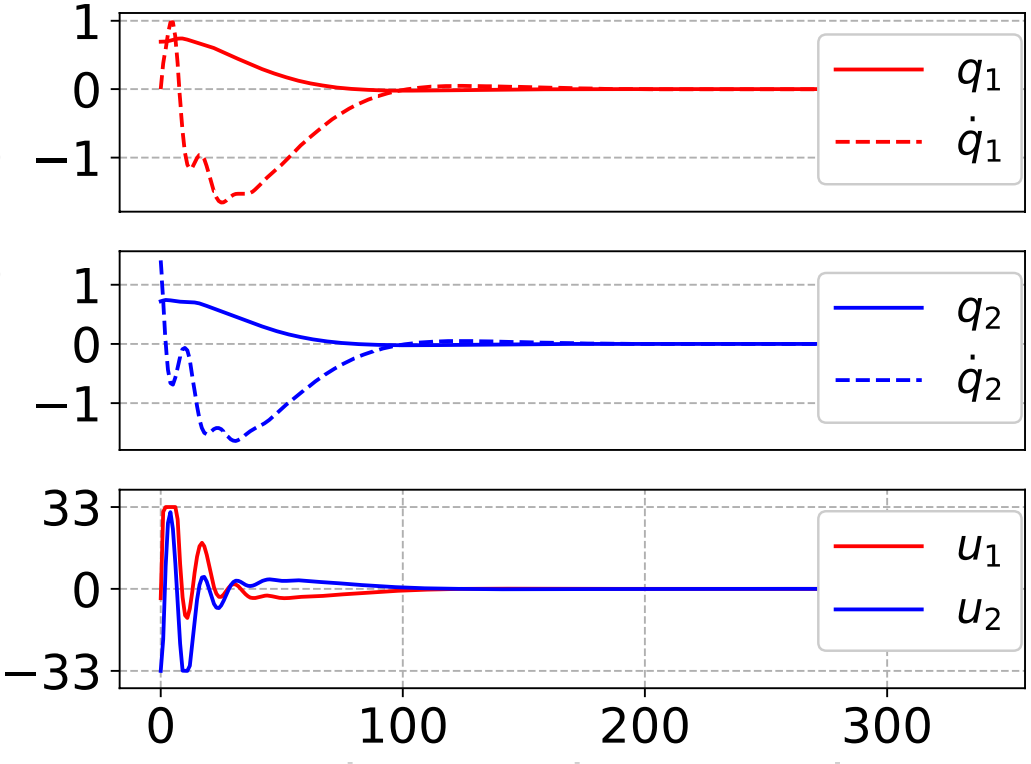
Example 2: position control in FPU system



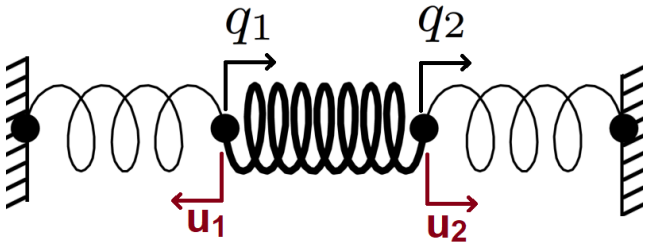
- Fermi-Pasta-Ulam chain with cubic spring potentials

$$\dot{x}_1(t) = x_3(t)$$
$$\dot{x}_2(t) = x_4(t)$$
$$\dot{x}_3(t) = -\frac{\eta^2}{2}\big(x_2(t) + x_1(t)\big) + 3x_1(t)^2 + u_1(t)$$
$$\dot{x}_4(t) = -\frac{\eta^2}{2}\big(x_2(t) + x_1(t)\big) + 3x_2(t)^2 + u_2(t)$$

- Forward Euler discretization, $\Delta = 10^{-2}$
  prediction horizon: $N = 25$
  constraints: $\|x_i(t)\|_\infty \leq 2$, $\|u_i(t)\|_\infty \leq 33$



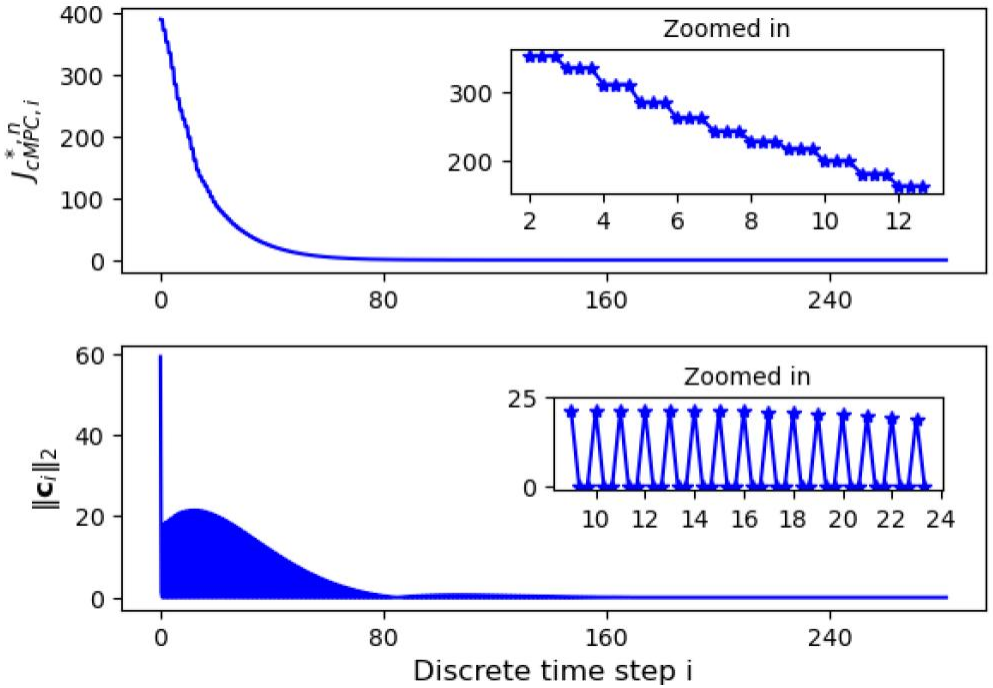
Example 2: position control in FPU system



- Fermi-Pasta-Ulam chain with cubic spring potentials

$$\dot{x}_1(t) = x_3(t)$$
$$\dot{x}_2(t) = x_4(t)$$
$$\dot{x}_3(t) = -\frac{\eta^2}{2}\big(x_2(t) + x_1(t)\big) + 3x_1(t)^2 + u_1(t)$$
$$\dot{x}_4(t) = -\frac{\eta^2}{2}\big(x_2(t) + x_1(t)\big) + 3x_2(t)^2 + u_2(t)$$

- Forward Euler discretization, $\Delta = 10^{-2}$
  prediction horizon: $N = 25$
  constraints: $\|x_i(t)\|_\infty \leq 2$, $\|u_i(t)\|_\infty \leq 33$



Example 2: computation time

	solver	maxiters	n_v	n_c	n_p	time for iter 1 $i=0$	**total time at** $i=0$	total time at $i=1$	total time at $i=2$	J_0^*	J_{cl}
Example 2	cvxpy	1	309	2034	854	0.970	1.284	0.192	0.192	386.77	198.85
	cvxpy	3				0.902	2.056	0.634	0.629	389.27	198.80
	cvxpy+jit	1				0.992	**1.259**	**0.167**	**0.152**	389.77	198.85
	cvxpy+jit	3				0.974	**1.608**	**0.447**	**0.459**	389.27	198.80
	NLP	-	154	709	-	-	31.602	34.365	37.811	389.27	198.80
	NLP+jit	-				-	**2.375**	**2.099**	**2.060**	389.27	198.80

- Sequential convex program (implemented via cvxpy) vs NLP (SLSQP scipy.optimize) with & without pre-compilation via jit (jax library)
- n_v, n_c, n_p: number of optimization variables, constraints, parameters
- Speedup factor: 14 $\times$ with jit, 197 $\times$ without jit

Robust DC-TMPC properties

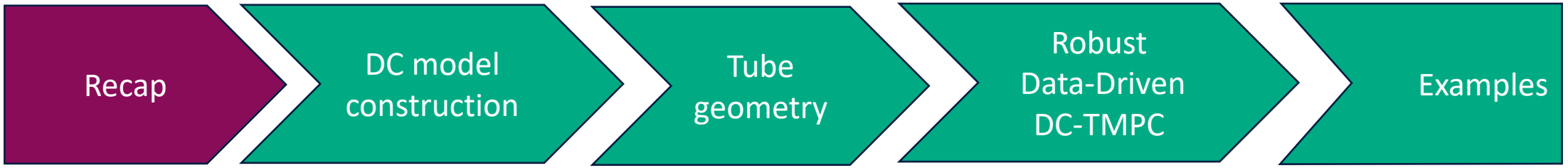
Guarantees:

- Convergence to a local solution of the original NLP
- Possibility of early termination with feasible solution
- Quadratic stability
- Robustness to additive disturbances
- Efficiency

Enabled by DC programing

Computing DC functions

- Assumption:
  $f(x, u) = g(x, u) - h(x, u)$, where $g, h : \mathcal{X} \times \mathcal{U} \to \mathbb{R}^{n_x}$ are differentiable and componentwise convex on the convex sets $\mathcal{X}, \mathcal{U}$
- Any $f \in \mathcal{C}^2$ can be expressed as $f = g - h$ for convex g, h
- Obtaining DC representations:
  - Analytic solutions are rare
  - Using polynomial approximations
  - Using ICNNs



Thank you!

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A successive convexification approach for robust receding horizon control
Y. Lishkova and M. Cannon. *IEEE Transactions on Automatic Control,* doi: 10.1109/TAC.2025.3558253 , 2025

Data-driven robust MPC of tiltwing VTOL aircraft
M. Doff-Sotta, M. Cannon, and M. Bacic. *Journal of Guidance, Control, and Dynamics*, doi:10.2514/1.G008315, 2024

Safe learning in nonlinear model predictive control
J. Buerger, M. Cannon, and M. Doff-Sotta. *6th Annual Conference on Learning for Dynamics and Control*, proceedings.mlr.press/v242/buerger24a, 2024



Handling nonlinearities and uncertainties of fed-batch cultivations with difference of convex functions tube MPC
N. Krausch, M. Doff-Sotta, M. Cannon, P. Neubauer, and M. N. Cruz Bournazou. *Computer Aided Chemical Engineering, Vol. 53, pp. 1699–1704,* doi:10.1016/B978-0-443-28824-1.50284-2, 2024

Difference of convex functions in robust tube nonlinear MPC
M. Doff-Sotta and M. Cannon. *61st IEEE Conference on Decision and Control,*doi:10.1109/CDC51059.2022.9993390, 2022

Robust TMPC

$$\min_{\boldsymbol{u}_i, \boldsymbol{X}_i} \left(\max_{x_{N|i}} \|x_{N|i} - x^r_{N+i}\|^2_{Q_N} + \right.$$

$$\left. \sum_{k=0}^{N-1} \max_{x_{k|i}} \left[\|x_{k|i} - x^r_{k+i}\|^2_Q + \|u_{k|i} - u^r_{k+i}\|^2_R \right] \right)$$

subject to, for $k = 0, \ldots, N-1$:

$$X_{k|i} \in \mathcal{X}$$

$$KX_{k|i} \oplus \{c_{k|i} + c^0_{k|i}\} \subseteq \mathcal{U},$$

$$[\bar{x}_{k+1|i}]_j \geq [f(x, Kx + c_{k|i} + c^0_{k|i})]_j + [\bar{w}]_j$$

$$[\underline{x}_{k+1|i}]_j \leq [f(x^0_{k|i,j}, u^0_{k|i,j})]_j +$$

$$[A_{k|i} + B_{k|i}K]_j (x - x^0_{k|i,j}) + [B_{k|i}]_j c_{k|i} + [\underline{w}]_j$$

for all $x \in X_{k|i}$, $j = 1, \ldots, n_x$

$$X_{0|i} = \{x_i\}$$

$$X_{N|i} \in \mathcal{X}_N \subseteq \mathcal{X}$$

$$x^0_{k|i,j} = \arg \min_{x \in X^0_{k|i}} [f(x, Kx + c^0_{k|i})]_j \quad (17)$$

$$[\bar{x}^0_{k+1|i}]_j = \max_{x \in X^0_{k|i}} [f(x, Kx + c^0_{k|i})]_j + [\bar{w}]_j$$

$$[\underline{x}^0_{k+1|i}]_j = [f(x^0_{k|i,j}, u^0_{k|i,j})]_j + [\underline{w}]_j \quad (19)$$

$$[\bar{w}]_j = \max_{w \in \mathcal{W}} [w]_j, \quad [\underline{w}]_j = \min_{w \in \mathcal{W}} [w]_j$$

$$X^0_{k|i+1} = \{x : \underline{x}^0_{k|i+1} \leq x \leq \bar{x}^0_{k|i+1}\}$$

Algorithm 2: Seed tube and linearization

Require: Initial state x_i and seed $\mathbf{c}^0_i$

1 Set $\bar{x}_{0|i} = \underline{x}_{0|i} = x_i$ so that $X^0_{0|i} \leftarrow \{x_i\}$;

2 **for** $k = 0, \ldots, N-1$ **do**

3 **for** $j = 1, \ldots, n_x$ **do**

4 Solve problem (17) for $x^0_{k|i,j}$ and $u^0_{k|i,j} = Kx^0_{k|i,j} + c^0_{k|i}$;

5 Compute $[A_{k|i}]_j = \frac{\partial [f]_j}{\partial x}\Big|_{(x^0_{k|i,j}, u^0_{k|i,j})}$ and $[B_{k|i}]_j = \frac{\partial [f]_j}{\partial u}\Big|_{(x^0_{k|i,j}, u^0_{k|i,j})}$;

6 Solve (18)-(19) for $[\bar{x}_{k+1|i}]_j$, $[\underline{x}_{k+1|i}]_j$;

7 Set $X^0_{k+1|i} \leftarrow \{x : \underline{x}_{k+1|i} \leq x \leq \bar{x}_{k+1|i}\}$;